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EXPERIMENT-5

PACKET CAPTURE USING WIRESHARK

AIM:

To capture filter inspect & analyze network protocols using the wireshark packet analyzer.

ALGORITHM:

→ Open Wireshark
→ Select the active network interface
→ Configure capture to stop after 100 packets
→ Start packet capture
→ Apply display filters
→ Inspect packets & view the flow graph
→ Save the captured packets.

PROCEDURE:

STEP 1: Launch wireshark & start the active network interface.

STEP 2: Go to capture → options & set stop capture after 100 packets.

STEP 3: Click start to design packet capture
→ TCP/UDP: tcp || udp
→ ARP: arp
→ DNS: dns
→ HTTP: http
→ DHCP: bootp

STEP 4: Inspect the filtered packets in the packet list, details & bytes.

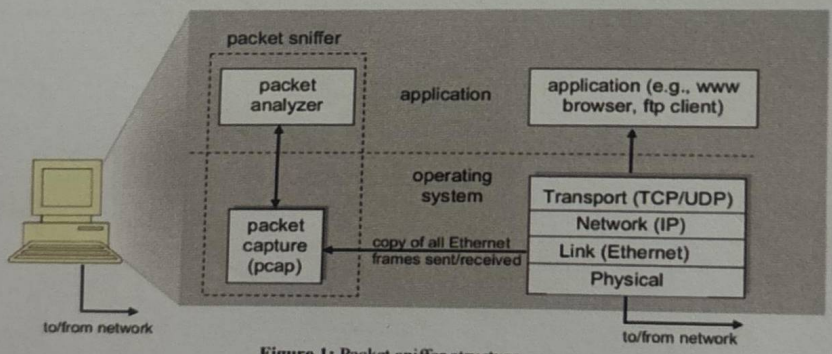
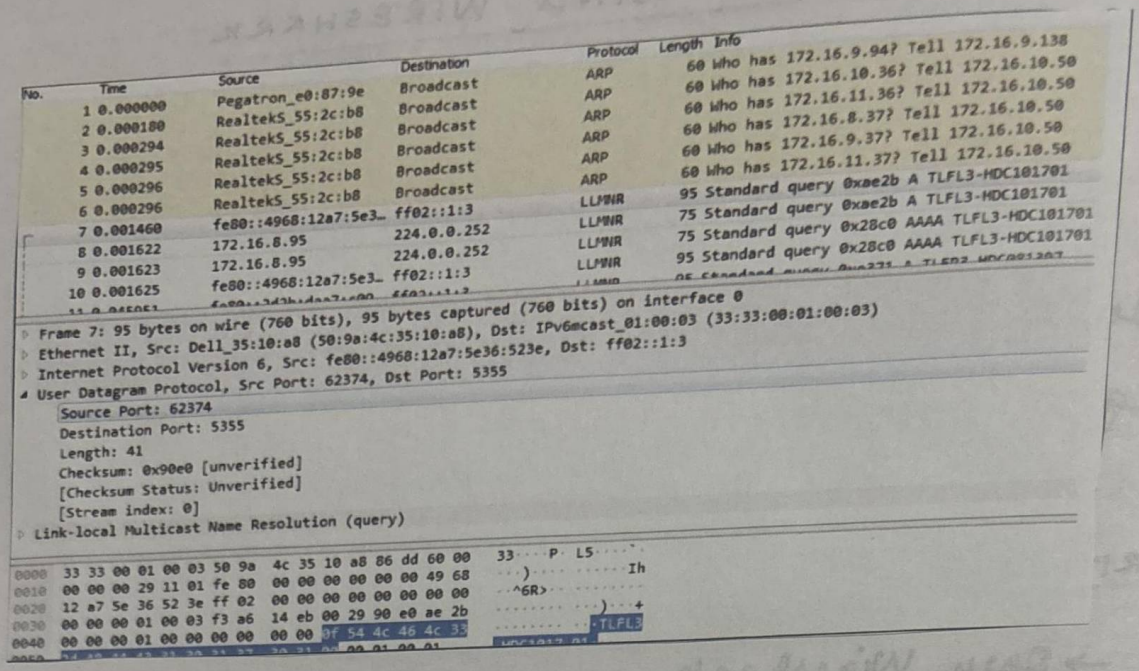


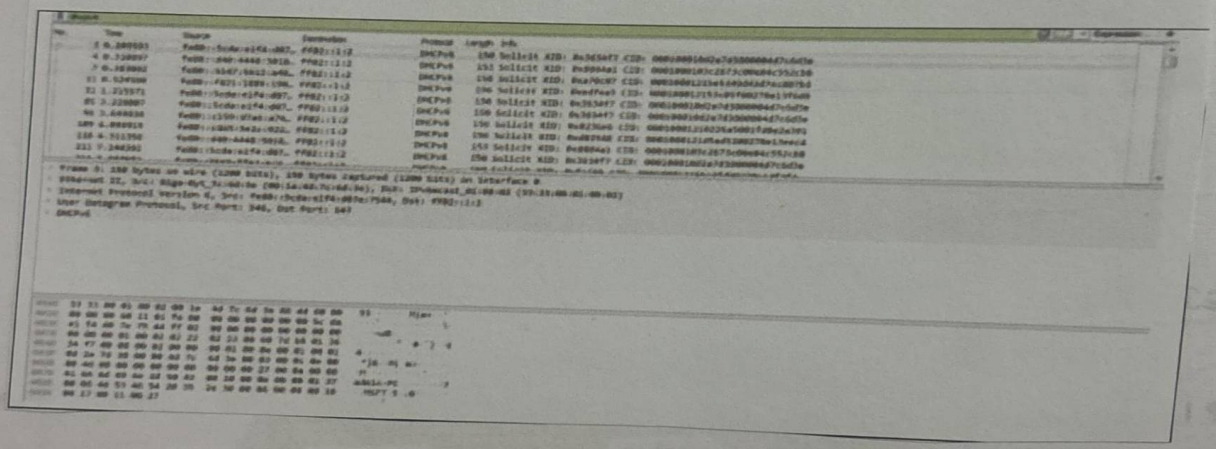
Figure 1: Packet sniffer structure

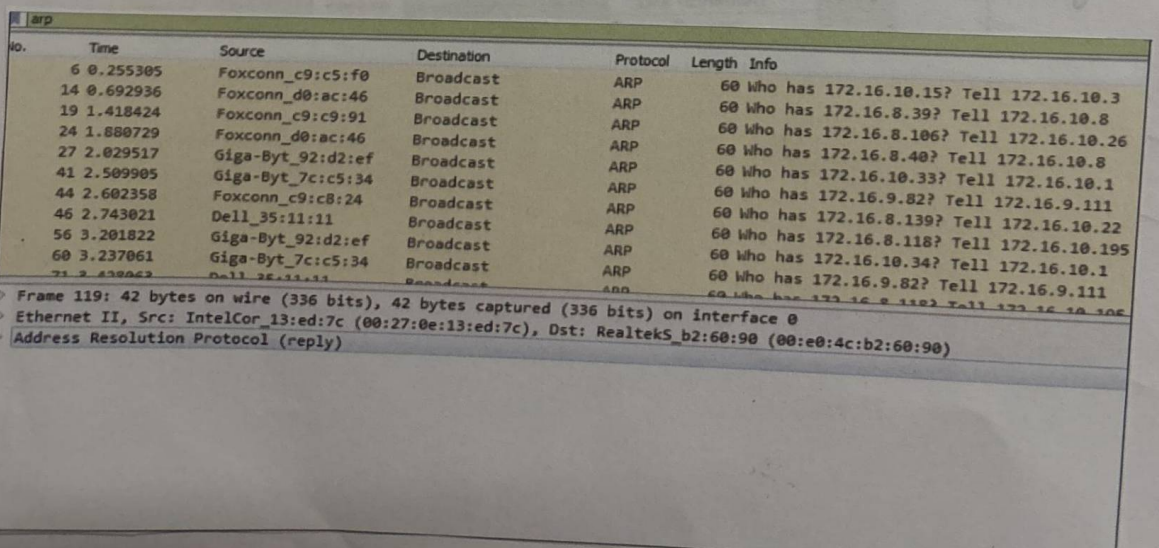


No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	Pegatron_e0:87:9e	Broadcast	ARP	60	Who has 172.16.9.94? Tell 172.16.9.138
2	0.000180	RealtekS_55:2c:b8	Broadcast	ARP	60	Who has 172.16.10.36? Tell 172.16.10.50
3	0.000294	RealtekS_55:2c:b8	Broadcast	ARP	60	Who has 172.16.11.36? Tell 172.16.10.50
4	0.000295	RealtekS_55:2c:b8	Broadcast	ARP	60	Who has 172.16.8.37? Tell 172.16.10.50
5	0.000296	RealtekS_55:2c:b8	Broadcast	ARP	60	Who has 172.16.9.37? Tell 172.16.10.50
6	0.000296	RealtekS_55:2c:b8	Broadcast	ARP	60	Who has 172.16.11.37? Tell 172.16.10.50
7	0.001460	fe80::4968:12a7:5e3…	ff02::1:3	LLMNR	95	Standard query 0xae2b A TLFL3-HDC101701
8	0.001622	172.16.8.95	224.0.0.252	LLMNR	75	Standard query 0xae2b A TLFL3-HDC101701
9	0.001623	172.16.8.95	224.0.0.252	LLMNR	75	Standard query 0x28c0 AAAA TLFL3-HDC101701
10	0.001625	fe80::4968:12a7:5e3…	ff02::1:3	LLMNR	95	Standard query 0x28c0 AAAA TLFL3-HDC101701

```
▷ Frame 7: 95 bytes on wire (760 bits), 95 bytes captured (760 bits) on interface 0
▷ Ethernet II, Src: Dell_35:10:a8 (50:9a:4c:35:10:a8), Dst: IPv6mcast_01:00:03 (33:33:00:01:00:03)
▷ Internet Protocol Version 6, Src: fe80::4968:12a7:5e36:523e, Dst: ff02::1:3
◢ User Datagram Protocol, Src Port: 62374, Dst Port: 5355
    Source Port: 62374
    Destination Port: 5355
    Length: 41
    Checksum: 0x90e0 [unverified]
    [Checksum Status: Unverified]
    [Stream index: 0]
▷ Link-local Multicast Name Resolution (query)
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```
0000  33 33 00 01 00 03 50 9a  4c 35 10 a8 86 dd 60 00   33····P· L5····`·
0010  00 00 00 29 11 01 fe 80  00 00 00 00 00 00 49 68   ···)···· ····Ih
0020  12 a7 5e 36 52 3e ff 02  00 00 00 00 00 00 00 00   ··^6R>·· ········
0030  00 00 00 01 00 03 f3 a6  14 eb 00 29 90 e0 ae 2b   ········ ···)···+
0040  00 00 00 01 00 00 00 00  00 00 0f 54 4c 46 4c 33   ········ ···TLFL3
```





arp

No.	Time	Source	Destination	Protocol	Length	Info
6	0.255305	Foxconn_c9:c5:f0	Broadcast	ARP	60	Who has 172.16.10.15? Tell 172.16.10.3
14	0.692936	Foxconn_d0:ac:46	Broadcast	ARP	60	Who has 172.16.8.39? Tell 172.16.10.8
19	1.418424	Foxconn_c9:c9:91	Broadcast	ARP	60	Who has 172.16.8.106? Tell 172.16.10.26
24	1.880729	Foxconn_d0:ac:46	Broadcast	ARP	60	Who has 172.16.8.40? Tell 172.16.10.8
27	2.029517	Giga-Byt_92:d2:ef	Broadcast	ARP	60	Who has 172.16.10.33? Tell 172.16.10.1
41	2.509905	Giga-Byt_7c:c5:34	Broadcast	ARP	60	Who has 172.16.9.82? Tell 172.16.9.111
44	2.602358	Foxconn_c9:c8:24	Broadcast	ARP	60	Who has 172.16.8.139? Tell 172.16.10.22
46	2.743021	Dell_35:11:11	Broadcast	ARP	60	Who has 172.16.8.118? Tell 172.16.10.195
56	3.201822	Giga-Byt_92:d2:ef	Broadcast	ARP	60	Who has 172.16.10.34? Tell 172.16.10.1
60	3.237061	Giga-Byt_7c:c5:34	Broadcast	ARP	60	Who has 172.16.9.82? Tell 172.16.9.111

```
▷ Frame 119: 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface 0
▷ Ethernet II, Src: IntelCor_13:ed:7c (00:27:0e:13:ed:7c), Dst: RealtekS_b2:60:90 (00:e0:4c:b2:60:90)
▷ Address Resolution Protocol (reply)
```

STEP 5: Save the capture file for future analysis.

RESULT:

Successfully captured 100 network packets, applied display filters for TCP, UDP, ARP, DNS, HTTP, IP, ICMP, DHCP are analyzed packet details using the wireshark.